

The non-divisor mass of $\binom{2n}{n}$ is bounded for almost every n

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Abstract

Let $E(n) = \sum_{p \nmid \binom{2n}{n}} 1/p$, so that the Erdős–Graham–Ruzsa–Straus conjecture is $E(n) = O(1)$. We prove, from Kummer’s carry criterion, the Chinese remainder theorem and Mertens’ theorem alone, that $E(n)$ is bounded for almost every n : for every $\lambda > 0$,

$$\#\{n \leq N : E(n) > 2.36 + \lambda\} \leq (0.21 + o(1)) N/\lambda^2,$$

and, driving the same mechanism with K rails instead of two, that the tail is in fact super-exponential:

$$\#\{n \leq N : E(n) > C\} \leq N \exp(-C \log \log C (1 + o(1))).$$

No equidistribution, no transversality, no Diophantine input, no ineffective constant. The mechanism is exact multi-rail registration, and the one thing that has to be got right is the pricing: resolving a prime p to depth d consumes arc length $d \log p$ of a carrier of length $\log N$, so depth must be allotted as a fraction of each rail’s *own* depth, not as an absolute number of digits. Under a uniform absolute depth the large-prime rails exhaust the budget and one recovers only $E(n) = O(\log \log \log n)$ off a set of size $N(\log N)^{-B}$; we record that variant too, since the two are the endpoints of a single trade governed by one free function — the depth profile.

1 Statement

For $n \geq 1$ put

$$E(n) = \sum_{\substack{p \leq n \\ p \nmid \binom{2n}{n}}} \frac{1}{p}.$$

Since $\sum_{p \leq n} 1/p = \log \log n + M + O(1/\log n)$, the Erdős–Graham–Ruzsa–Straus conjecture $\sum_{p \nmid \binom{2n}{n}} 1/p = (1 + o(1)) \log \log n$ is the statement $E(n) = o(\log \log n)$, and the full registry problem (no. 377) is $E(n) = O(1)$.

Theorem 1 (Bounded for almost every n). *For every $\lambda > 0$ and all sufficiently large N ,*

$$\#\{n \leq N : E(n) > 2.36 + \lambda\} \leq (0.21 + o(1)) \frac{N}{\lambda^2}.$$

In particular $E(n) \leq 4.36$ for all but 5.1% of $n \leq N$, and $E(n) \leq 12.36$ for all but 0.21%.

Theorem 2 (Super-exponential tail). *There are absolute constants $c < 1$ and K_0 such that for every integer $K \geq K_0$,*

$$\#\{n \leq N : E(n) > K + \log K + 1\} \leq N \left(\frac{c}{\log K} \right)^K + o(N).$$

Equivalently, for every C large,

$$\#\{n \leq N : E(n) > C\} \leq N \exp(-C \log \log C (1 + o(1))).$$

Theorem 3 (Deep-rail variant). *There is an absolute constant C_0 such that for all large N and every integer k with $3 \leq k \leq \frac{1}{2} \log N$,*

$$\#\{n \leq N : E(n) > \log k + C_0\} \leq N (\log \log N) \left(\frac{2}{3} \right)^{\lfloor k/2 \rfloor}.$$

Consequently, for every $B \geq 1$, all but $O_B(N(\log N)^{-B})$ integers $n \leq N$ satisfy $E(n) \leq \log \log N + O(\log B)$.

Theorem 1 is the stronger statement about the bound; Theorem 3 the stronger statement about the exceptional set. §6 shows that they are the two ends of one trade, and identifies the free function that governs it.

2 The three inputs

(K) Kummer's carry criterion. For an odd prime p , $v_p\binom{2n}{n}$ is the number of carries when $n + n$ is added in base p . A carry leaves digit position i exactly when $2a_i + \varepsilon_i \geq p$, with $\varepsilon_i \in \{0, 1\}$ the incoming carry; inductively there are no carries at all iff $2a_i < p$ for every i . Hence, with

$$D_p := \left\{0, 1, \dots, \frac{p-1}{2}\right\}, \quad |D_p| = \frac{p+1}{2},$$

$$p \nmid \binom{2n}{n} \iff \text{every base-}p \text{ digit of } n \text{ lies in } D_p. \quad (1)$$

For $p = 2$ one has $v_2\binom{2n}{n} = s_2(n) \geq 1$ for $n \geq 1$, so 2 never contributes to $E(n)$ and is excluded throughout.

(R) Rail truncation. For $d \geq 1$ let

$$X_p^{(d)}(n) = \mathbf{1}[a_0, \dots, a_{d-1} \in D_p],$$

a_i the base- p digits of n . By (1), $\mathbf{1}[p \nmid \binom{2n}{n}] \leq X_p^{(d)}(n)$ for every $d \geq 1$ and every n . Crucially $X_p^{(d)}$ depends only on $n \bmod p^d$, and

$$\rho_p^{(d)} := \frac{|D_p|^d}{p^d} = \left(\frac{1}{2} + \frac{1}{2p} \right)^d \leq \left(\frac{2}{3} \right)^d \quad (p \geq 3). \quad (2)$$

(C) **Exact registration.** The single fact that replaces all analytic input.

Lemma 4 (Rail registration). *Let $p_1, \dots, p_K$ be distinct odd primes, $d_1, \dots, d_K \geq 1$, and $M = \prod_i p_i^{d_i}$. Then*

$$\frac{1}{N} \sum_{n \leq N} \prod_{i=1}^K X_{p_i}^{(d_i)}(n) = \left(\prod_{i=1}^K \rho_{p_i}^{(d_i)} \right) \left(1 + O(M/N) \right),$$

with an absolute implied constant. The relative error vanishes identically when $M \mid N$.

Proof. Each $X_{p_i}^{(d_i)}$ depends only on $n \bmod p_i^{d_i}$, and the moduli $p_i^{d_i}$ are pairwise coprime, so the Chinese remainder theorem gives a ring isomorphism $\mathbb{Z}/M \cong \prod_i \mathbb{Z}/p_i^{d_i}$ under which the joint condition is a product of sets. The admissible residues mod M therefore number exactly $R = \prod_i |D_{p_i}^{d_i}|$, and counting $n \leq N$ in a union of R classes mod M gives $R \lfloor N/M \rfloor + \theta R = RN/M + O(R)$ with $|\theta| \leq 1$. Finally $R/M = \prod_i \rho_{p_i}^{(d_i)}$, so the absolute error $O(R)$ is $O(M/N)$ relative to the main term NR/M . $\square$

Remark 5 (Why no analysis appears). *Lemma 4 is the entire comparison mechanism, and it is an identity, not an estimate. Harmonically: the depth- d condition on rail p has all its Fourier support in $p^{-d}\mathbb{Z}$; two rails' mode lattices $p^{-d}\mathbb{Z}$ and $q^{-e}\mathbb{Z}$ meet only in $\mathbb{Z}$; every nontrivial cross mode therefore cancels exactly, because coprimality is an exact algebraic fact and not an approximation. The familiar question “how close is a $\log p$ to $b \log q$ ” — and with it the irrationality measure of $\log p / \log q$, Baker's theorem, and the ineffective constants they carry — belongs to the real-scale readout of these sets as self-similar subsets of $[0, 1]$. It does not arise here.*

Bands. For $p \leq N$ put $k_p := \lfloor \log N / \log p \rfloor$, so that band k is $(N^{1/(k+1)}, N^{1/k}]$ and, by Mertens,

$$\sum_{p \in \text{band } k} \frac{1}{p} = \log \left(1 + \frac{1}{k} \right) + o(1), \quad \sum_{p > N^{1/k_1}} \frac{1}{p} = \log k_1 + o(1), \quad (3)$$

the second by telescoping $\sum_{k < k_1} \log(1 + 1/k) = \log k_1$.

3 Proof of Theorem 1

The depth profile. This is the only design choice in the proof, and the whole theorem turns on it. Set

$$k_1 := 8, \quad d_p := \lceil k_p / 4 \rceil \quad \text{for } p \leq N^{1/8}, \quad (4)$$

so each rail is resolved to a fixed *fraction* of its own depth rather than to a fixed number of digits. Since $p \leq N^{1/k_p}$ and $k_p \geq 8$,

$$p^{d_p} \leq N^{d_p/k_p} \leq N^{\frac{1}{4} + \frac{1}{k_p}} \leq N^{3/8}. \quad (5)$$

Every rail therefore costs at most $N^{3/8}$, uniformly in p ; this is what makes pairs affordable, and it fails badly for a uniform absolute depth (§6).

Step 1: split off the unresolved bands. By (1), (R) and (3),

$$E(n) \leq \underbrace{\sum_{p > N^{1/8}} \frac{1}{p}}_{= \log 8 + o(1)} + F(n), \quad F(n) := \sum_{3 \leq p \leq N^{1/8}} \frac{1}{p} X_p^{(d_p)}(n). \quad (6)$$

Step 2: the mean. By Lemma 4 with $K = 1$ and (5), $\frac{1}{N} \sum_{n \leq N} X_p^{(d_p)} = \rho_p^{(d_p)}(1 + O(N^{-5/8}))$, so by (2) and (3)

$$\mu := \frac{1}{N} \sum_{n \leq N} F(n) = \sum_{k \geq 8} \log \left(1 + \frac{1}{k} \right) \left(\frac{2}{3} \right)^{\lceil k/4 \rceil} + o(1) \leq 0.2787 + o(1), \quad (7)$$

the numerical value by direct summation of the (geometrically convergent) series.

Step 3: the variance, by exact pair registration. For distinct $p, q \leq N^{1/8}$ we have $p^{d_p} q^{d_q} \leq N^{3/4}$ by (5), so Lemma 4 with $K = 2$ gives

$$\frac{1}{N} \sum_{n \leq N} X_p^{(d_p)} X_q^{(d_q)} = \rho_p^{(d_p)} \rho_q^{(d_q)} (1 + O(N^{-1/4})),$$

i.e. the two rails are uncorrelated up to $O(N^{-1/4})$. Hence

$$\text{Var}(F) \leq \sum_{3 \leq p \leq N^{1/8}} \frac{\rho_p^{(d_p)}}{p^2} + O(N^{-1/4}) \leq \sum_{p \geq 3} \frac{1}{p^2} + o(1) = 0.2023 + o(1), \quad (8)$$

using $\sum_p p^{-2} = 0.45224 \dots$ and subtracting the term $p = 2$.

Step 4: Chebyshev. By (7) and (8), $\#\{n \leq N : F(n) > \mu + \lambda\} \leq (0.2023 + o(1))N/\lambda^2$. Combining with (6), and since $\log 8 + 0.2787 = 2.3581$, any n with $E(n) > 2.36 + \lambda$ has $F(n) > \mu + \lambda$ for large N . Theorem 1 follows. $\square$

4 Proof of Theorem 3

Here a single rail is driven deep instead. Fix $3 \leq k \leq \frac{1}{2} \log N$ and put $d := \lfloor k/2 \rfloor$, $P_0 := N^{1/k}$. As in (6),

$$E(n) \leq \sum_{p > P_0} \frac{1}{p} + F_k(n) = \log k + O(1) + F_k(n), \quad F_k(n) := \sum_{3 \leq p \leq P_0} \frac{1}{p} X_p^{(d)}(n),$$

by (3). For $p \leq P_0$ one has $p^d \leq P_0^{k/2} = N^{1/2}$, so Lemma 4 with $K = 1$ applies and, by (2) and Mertens,

$$\frac{1}{N} \sum_{n \leq N} F_k(n) \leq \left(\frac{2}{3} \right)^d (\log \log N + O(1)).$$

Since $F_k \geq 0$, Markov's inequality gives $\#\{n \leq N : F_k(n) > 1\} \leq N(\log \log N + O(1))(2/3)^d$, and absorbing the $O(1)$'s into C_0 proves the first display. For the consequence take $k = \lceil 6(B+1) \log \log N \rceil$: then $(2/3)^{\lfloor k/2 \rfloor} \ll (\log N)^{-1.21(B+1)}$, so the exceptional count is $\ll N(\log \log N)(\log N)^{-1.21(B+1)} \ll N(\log N)^{-B}$, while $\log k = \log \log \log N + O(\log B)$. $\square$

5 Proof of Theorem 2

Fix $K \geq K_0$ and warp with $\theta = 1/(2K)$: discard the bands $k < k_1 := 2K$ and set $d_p := \lceil k_p/(2K) \rceil$ for $p \leq N^{1/2K}$. Then $d_p/k_p \leq 1/(2K) + 1/k_1 = 1/K$, so any K -tuple of such rails costs $M \leq N^{K \cdot 1/K}$; replacing θ by $\theta/2$ makes this $M \leq N^{1/2}$, and Lemma 4 applies to every K -tuple. As in (6),

$$E(n) \leq \log k_1 + o(1) + F(n) = \log K + \log 2 + o(1) + F(n), \quad F(n) = \sum_{3 \leq p \leq N^{1/2K}} \frac{1}{p} X_p^{(d_p)}(n),$$

and $\mu := \frac{1}{N} \sum_{n \leq N} F(n) \leq \sum_{k \geq k_1} \log(1 + \frac{1}{k}) (\frac{2}{3})^{\lceil \theta k \rceil} \leq \int_1^\infty (\frac{2}{3})^v \frac{dv}{v} + o(1) < 0.71$, uniformly in K .

The K -th moment. Write $F = \sum_p a_p Y_p$ with $a_p = 1/p \leq A := \frac{1}{3}$ and $Y_p := X_p^{(d_p)}$. Expanding F^K and grouping the K -tuples by the partition of $\{1, \dots, K\}$ they induce (equal indices in the same block), Lemma 4 gives, for each partition into blocks of sizes $m_1, \dots, m_j$,

$$\frac{1}{N} \sum_{n \leq N} \sum_{q_1, \dots, q_j \text{ distinct}} \prod_l a_{q_l}^{m_l} \prod_l Y_{q_l} \leq \prod_{l=1}^j \left(\sum_q a_q^{m_l} \rho_q \right) (1 + o(1)) \leq \mu^j A^{K-j} (1 + o(1)),$$

since $\sum_q a_q^{m_l} \rho_q \leq A^{m-1} \mu$. Summing over partitions, with $S(K, j)$ the Stirling numbers of the second kind and $T_K(x) = \sum_j S(K, j) x^j$ the Touchard polynomial,

$$\frac{1}{N} \sum_{n \leq N} F(n)^K \leq A^K T_K(\mu/A) (1 + o(1)), \quad T_K(x) \leq \left(\frac{K}{\log(1 + K/x)} \right)^K,$$

the last a standard bound ($T_K(x) = \mathbb{E} \text{Poi}(x)^K$; for $x = 1$ it is the Berend–Tassa bound on Bell numbers). With $\mu/A < 2.13$,

$$\frac{1}{N} \#\{n \leq N : F(n) > K\} \leq \frac{A^K T_K(\mu/A)}{K^K} (1 + o(1)) \leq \left(\frac{1}{3 \log(K/2.13)} \right)^K (1 + o(1)),$$

which is $\leq (c/\log K)^K$ for an absolute $c < 1$ and $K \geq K_0$. Since $E(n) \leq \log K + \log 2 + o(1) + F(n)$, any n with $E(n) > K + \log K + 1$ has $F(n) > K$, giving the first display. For the second, given C take $K = \lceil C - \log C - 1 \rceil$, so that $K + \log K + 1 \leq C$ and $(c/\log K)^K = \exp(-K \log(\log K/c)) = \exp(-C \log \log C(1 + o(1)))$. $\square$

6 Scaling, warping, and the trade

The budget is an arc length. Resolving rail p to depth d consumes modulus p^d , i.e. arc length $d \log p$ on a carrier of length $\log N$. Lemma 4 is informative for a K -tuple exactly when

$$\sum_{i=1}^K d_i \log p_i \leq (1 - \eta) \log N, \tag{9}$$

which in band coordinates ($p_i \approx N^{1/k_i}$, full depth $d_i = k_i$) reads $\sum_i d_i/k_i \leq 1 - \eta$: *the total resolution of all rails in a tuple, each measured in units of its own full depth, cannot exceed one carrier length.*

Scaling versus warping. Each rail carries a constant scale, $\log p$ — that is the scaling, and it is fixed by the rail. What is free is the *depth profile* $d(k) = \lceil \theta(k) k \rceil$, a rescaling function laid along the carrier: a warp. The two layer, and the layering is where the theorem lives. A uniform absolute depth $d(k) \equiv d$ is a pure scaling with no warp, and it prices every rail in the chart's unit rather than its own: one level of a rail at $p \approx \sqrt{N}$ then costs half the carrier, so all bands with $k < 2d$ must be discarded and one pays their entire Mertens mass $\log(2d)$. That is exactly the $\log k$ term in Theorem 3, and it is what forces the $\log \log \log N$ floor there — an artifact of the pricing, not of the problem. Under the warp (4) every rail costs $N^{\theta+1/k}$ regardless of p , the discard threshold k_1 is decoupled from the depth, and the floor disappears: Theorem 1.

The trade. With profile θ and K rails, (9) demands $K\theta + \sum_i 1/k_i \leq 1 - \eta$, so the discard threshold must satisfy $k_1 \gtrsim K$ and one pays $\log k_1 \approx \log K$; the resolved mean stays $O(1)$ provided $\theta k_1 \gtrsim 1$, and a K -th moment gives an exceptional set $\ll N(C/\lambda)^K$. Hence

$$E(n) \leq \log K + O(1) + \lambda \quad \text{off} \quad \ll N(C/\lambda)^K.$$

$K = 2$ is Theorem 1 (bound $O(1)$, exceptional N/λ^2); $K \asymp B \log \log N$ is Theorem 3 (bound $\log \log \log N$, exceptional $N(\log N)^{-B}$). Both ends are proved above from the first and second moments only.

Where the profile can still be pushed. Under the accounting (9) the binding tuple is K rails from the cheapest band, which forces $\sup \theta \leq (1 - \eta)/K$ and makes the constant profile optimal. That accounting is however not forced: two rails resolved on *disjoint height windows* of the carrier are independent up to e^{-g} , g the gap between the windows, rather than up to M/N . Prices under the window accounting are additive in arc length but insensitive to which rail spends where, which is a genuinely different optimization and the natural next lever. We state this as program, not as result.

Remark 6 (Register). *Theorems 1 and 3 are proven above with absolute implied constants; the numerical values 0.2787, 0.2023, $\log 8$ are explicit. §6's final paragraph is program. Measured: over 400 random $n \in [10^3, 2 \cdot 10^5]$, $\max E(n) = 0.869$, median 0.457, mean 0.472 — consistent with, and far inside, Theorem 1. **The literature gate has not been run:** the argument is elementary and an almost-all bound of this shape may be known. Erdős–Graham–Ruzsa–Straus (1975) and the subsequent literature on $\binom{2n}{n}$ must be read at source, with pin-cites, before any novelty claim is made. What is asserted without qualification is only that the statements are true and the proofs complete.*